

Human Resource Management System (HRMS)

Project Documentation

1. Introduction

The Human Resource Management System (HRMS) is a fully developed web-based application designed to digitize and automate core human resource operations within an organization. The system centralizes employee records, attendance tracking, leave management, payroll processing, and project/task coordination into a single secure platform.

2. Purpose of the System

The purpose of the HRMS is to reduce manual HR workload, improve operational transparency, ensure accurate attendance and payroll processing, and provide employees with self-service capabilities. The system enhances efficiency, accountability, and communication between administrators and employees.

3. Overall System Description

The HRMS operates with role-based access control, primarily divided into two user roles: Administrator and Employee. Each role has dedicated functionalities tailored to its responsibilities. The system ensures secure authentication and controlled access to sensitive organizational data.

4. Administrator Functionalities

- **Dashboard:** Displays organizational statistics including total employees, attendance summaries, leave statistics, and recent system activities.
- **Employee Management:** Create, view, update, and delete employee records. Manage employee credentials and role assignments.

- **Attendance Monitoring:** Track daily check-ins and check-outs in real time. Implements a 15-minute grace period for late arrivals and automatic absence marking when no attendance is recorded.
- **Leave Management:** Review, approve, or reject leave requests submitted by employees. Special absence weighting logic is applied for Saturday and Monday (x3 rule if configured).
- **Payroll Management:** Calculate salaries based on attendance and leave records. Generate structured payslips for employees.
- **Project & Task Management:** Create projects, assign teams, and allocate tasks. Tasks are privacy-focused, meaning only assigned employees can view their tasks.
- **System Settings:** Configure system-level rules and operational parameters.

5. Employee Functionalities

- **Personal Dashboard:** Overview of attendance status, assigned tasks, and important updates.
- **Attendance Marking:** Employees can check-in and check-out daily through the system.
- **Leave Requests:** Submit leave applications and monitor approval status.
- **Task Management:** View assigned tasks privately, update status, and manage priorities.
- **Profile Management:** Update personal information such as contact details securely.
- **Project Access:** View projects they are assigned to and related responsibilities.

6. Attendance Management Logic

The system includes automated attendance logic. A 15-minute grace period is applied to standard working hours for late arrivals. If an employee fails to mark attendance, the system can automatically mark them as absent. Special logic may apply to consecutive absences around weekends (Saturday/Monday rule) depending on configuration.

7. Payroll Processing

Payroll calculations are generated based on attendance records, approved leaves, and defined salary structures. The system ensures accurate salary computation and enables structured payslip generation for transparency.

8. Security Features

The HRMS implements secure authentication using JSON Web Tokens (JWT). User passwords are encrypted using hashing mechanisms. Role-based access control ensures that users can only access features relevant to their role.

9. Technology Stack Overview

The system is built using modern web technologies. The frontend utilizes React with Vite and Tailwind CSS for responsive design. The backend is developed using Node.js and Express, with MongoDB as the database managed through Mongoose. Additional libraries support authentication, date management, and notifications.

10. Conclusion

The HRMS is a comprehensive, secure, and scalable solution that automates core HR processes. It enhances operational efficiency, ensures data accuracy, and provides a structured platform for both administrative control and employee self-service.